

12.5.9
by 17.5

Тренировка 25.03.2026

12

$$Z = 2x_1 + x_2 - x_3 - x_4 \rightarrow \min$$

$$\begin{cases} x_1 + x_2 + 2x_3 + x_4 = 2 \\ 2x_1 + x_2 + 3x_3 + x_4 = 6 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases}$$

$$-Z = -2x_1 - x_2 + x_3 + x_4 - M(x_5 + x_6) \rightarrow \max$$

$$\begin{cases} x_1 + x_2 + 2x_3 + x_4 + x_5 = 2 \\ 2x_1 + x_2 + 3x_3 + x_4 + x_6 = 6 \\ x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{cases}$$

$$x_5 = 2 - x_1 - x_2 - 2x_3 - x_4$$

$$x_6 = 6 - 2x_1 - x_2 - 3x_3 - x_4$$

$$-Z = -2x_1 - x_2 + x_3 + x_4 - M(8 - 3x_1 - 2x_2 - 5x_3 - 2x_4)$$

Шаг 1:

б.к		x_1	x_2	x_3	x_4	x_5	x_6	св
x_5	2	1	1	2	1	1	0	1
x_6	6	2	1	3	1	0	1	2
$-Z$	0	2	1	1	-1	0	0	
$-M$	-8	-3	-2	-5	-2	0	0	

$$-Z = -2 \cdot 0 - 0 + 0 + 0 - M(8 - 3 \cdot 0 - 2 \cdot 0 - 5 \cdot 0 - 2 \cdot 0) = 0 - 8M$$

$$Z(0; 0; 0; 0; 2; 6)$$

Problem 2:

δ, R	1	x_1	x_2	x_3	x_4	x_5	x_6	CD
x_3	1	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	2
x_6	3	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{3}{2}$	1	6
-2	-1	$\frac{3}{2}$	$\frac{1}{2}$	0	$-\frac{3}{2}$	$-\frac{1}{2}$	0	
-14	-3	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{5}{2}$	0	

$$6 - \frac{2 \cdot 3}{2} = 6 - 3 = 3$$

$$0 - \frac{2 \cdot 1}{2} = -1 \quad \bigg| \quad -8 - \frac{(2 \cdot -5)}{2} = -3$$

$$2 - \frac{1 \cdot 3}{2} = 2 - \frac{3}{2} = \frac{4}{2} - \frac{3}{2} = \frac{1}{2}$$

$$1 - \frac{1 \cdot 3}{2} = \frac{2}{2} - \frac{3}{2} = -\frac{1}{2}$$

$$2 - \frac{1 \cdot 1}{2} = 2 - \frac{1}{2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}$$

$$-3 - \frac{(1 \cdot -5)}{2} = -3 + \frac{5}{2} = -\frac{6}{2} + \frac{5}{2} = -\frac{1}{2}$$

$$1 - \frac{1 \cdot 1}{2} = 1 - \frac{1}{2} = \frac{2}{2} - \frac{1}{2} = \frac{1}{2}$$

$$-2 - \frac{(1 \cdot -5)}{2} = -2 + \frac{5}{2} = -\frac{4}{2} + \frac{5}{2} = \frac{1}{2}$$

$$1 - \frac{1 \cdot 3}{2} = 1 - \frac{3}{2} = -\frac{1}{2}$$

$$-1 - \frac{1 \cdot 1}{2} = -1 - \frac{1}{2} = -\frac{2}{2} - \frac{1}{2} = -\frac{3}{2}$$

$$-2 - \frac{(1 \cdot -5)}{2} = -2 + \frac{5}{2} = -\frac{4}{2} + \frac{5}{2} = \frac{1}{2}$$

$$0 - \frac{1 \cdot 3}{2} = -\frac{3}{2}$$

$$0 - \frac{1 \cdot 1}{2} = -\frac{1}{2}$$

$$0 - \frac{1 \cdot (1 - 5)}{2} = \frac{5}{2}$$

Задача 3:

Л.П.	1	x_1	x_2	x_3	x_4	x_5	x_6
x_1	2	1	1	2	1	1	0
x_2	2	0	-1	-1	-1	1	1
$-z$	-4	0	-2	-3	-3	1	0
$-M$	-2	0	1	1	1	1	0

$$3 - 1 \cdot \frac{20}{2} = 3 - 10 = -7$$

$$-1 - 1 \cdot \frac{3}{2} = -1 - 1.5 = -2.5$$

$$-3 - 1 \cdot \frac{1}{2} = -3 - 0.5 = -3.5$$

$$-\frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{2} - \frac{1}{4} = -\frac{3}{4}$$

$$\frac{1}{2} - \frac{1}{2} \cdot \frac{-1}{2} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$0 - 1 \cdot \frac{1}{2} = -\frac{1}{2}$$

$$0 - 1 \cdot \frac{3}{2} = -\frac{3}{2}$$

$$0 - 1 \cdot \frac{1}{2} = -\frac{1}{2}$$

$$\frac{1}{2} - \frac{1}{2} \cdot -\frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$$

$$-\frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = -1$$

$$-\frac{3}{2} - \frac{1}{2} \cdot \frac{3}{2} = -\frac{3}{2} - \frac{3}{2} = -\frac{6}{2} = -3$$

Problem: unbounded ¹² ~~optimal~~ solution.

$$Z = 2x_1 - x_2 - x_3 - x_4 \rightarrow \max$$

$$\begin{cases} x_1 + x_2 + 2x_3 - x_4 = 2 \\ 2x_1 + x_2 - 3x_3 + x_4 = 6 \\ x_1 + x_2 + x_3 + x_4 = 7 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases} \quad \left(\begin{array}{cccc|c} 1 & 1 & 2 & -1 & 2 \\ 2 & 1 & -3 & 1 & 6 \\ 1 & 1 & 1 & 1 & 7 \end{array} \right)$$

$$\begin{cases} x_1 + x_2 + 2x_3 - x_4 + x_5 = 2 \\ 2x_1 + x_2 - 3x_3 + x_4 + x_6 = 6 \\ x_1 + x_2 + x_3 + x_4 - x_7 = 7 \\ x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0 \end{cases}$$

$$\cancel{Z = 2x_1 - x_2 - x_3 - x_4}$$

$$\begin{cases} x_5 = 2 - x_1 - x_2 - 2x_3 + x_4 \\ x_6 = 6 - 2x_1 - x_2 + 3x_3 - x_4 \\ x_7 = 7 - x_1 - x_2 - x_3 - x_4 \end{cases}$$

$$\begin{aligned} \hat{Z} &= 2x_1 - x_2 - x_3 - x_4 - M(x_5 + x_6 + x_7) = \\ &= 2x_1 - x_2 - x_3 - x_4 - M(15 - 4x_1 - 3x_2 - x_4) \end{aligned}$$

Шаг 1:

б.к	1	x_1	x_2	x_3	x_4	x_5	x_6	x_7	CO
x_1	1	1	1	2	-1	1	0	0	2
x_5	2	1	1	-3	1	0	1	0	3
x_6	6	2	1	1	1	0	0	1	7
x_7	7	1	1	1	1	0	0	0	11
Σ	0	-2	1	1	1	0	0	0	11
$-M$	-15	-4	-3	0	-1	0	0	0	11

$$Z = Z(0; 0; 0; 0; 2; 0; 7) =$$

$$= 2 \cdot 0 - 0 - 0 - 0 - 0 - M(15 - 4 \cdot 0 - 3 \cdot 0 - 0) =$$

$$= 0 - 15M$$

Шаг 2:

б.к	1	x_1	x_2	x_3	x_4	x_5	x_6	x_7	CO
x_1	2	1	1	2	-1		0	0	-
x_6	6	0	-1	-7	3		1	0	$\frac{2}{3}$
x_7	5	0	0	-1	2		0	1	$\frac{5}{2}$
Σ	4	0	3	5	-1		0	0	
$-M$	-7	0	1	8	-5		0	0	

$$6 - \frac{2 \cdot 2}{1} = 6 - 4 = 2$$

$$7 - \frac{2 \cdot 1}{1} = 7 - 2 = 5$$

$$0 - \frac{2 \cdot (-2)}{1} = 4$$

$$-15 - \frac{(2 \cdot -4)}{1} = -15 + 8 = -7$$

$$11 - \frac{1 \cdot 2}{1} = 11 - 2 = 9$$

$$4 - \frac{1 \cdot 1}{1} = 3$$

$$11 - \frac{1 \cdot -2}{1} = 11 + 2 = 13$$

$$0 - \frac{1 \cdot (-3)}{1} = 3$$

$$-3 - \frac{2 \cdot 2}{1} = -3 - 4 = -7$$

$$1 - \frac{1 \cdot 2}{1} = 1 - 2 = -1$$

$$1 - \frac{(-5) \cdot 2}{1} = 1 + 10 = 11$$

$$-3 - \frac{-4 \cdot 1}{1} = -3 + 4 = 1$$

$$1 - \frac{2 \cdot -1}{1} = 1 + 2 = 3$$

$$1 - \frac{2 \cdot 1}{1} = 1 - 2 = -1$$

$$1 - \frac{(-1) \cdot 13}{1} = 1 + 13 = 14$$

$$1 - 2 = -1$$

$$-1 - \frac{(-1) \cdot -4}{1} = -1 - 4 = -5$$

$$Z = Z(2; 0; 0; 0; 0; 2; 5) =$$

$$= 2 \cdot 2 - 0 - 0 - 0 - 0 - 11(15 - 1 \cdot 2 - 3 \cdot 0 - 0) = 4 - 711$$

Таблица 3:

$$-1 - \frac{-2 \cdot 2}{3} = -1 + \frac{4}{3} = \frac{1}{3}$$

с.л.	1	x_1	x_2	x_3	x_4	x_5	x_6	x_7	св
x_1	2	1	$\frac{2}{3}$	$-\frac{1}{3}$	0			0	-
x_4	$\frac{2}{3}$	0	$-\frac{1}{3}$	$-\frac{2}{3}$	1			0	-
x_3	$\frac{11}{3}$	0	$\frac{2}{3}$	$\frac{14}{3}$	0			1	1
$\frac{11}{3}$	$\frac{14}{3}$	0	$\frac{10}{3}$	$\frac{8}{3}$	0			0	-
-11	$-\frac{11}{3}$	0	$-\frac{2}{3}$	$-\frac{14}{3}$	0			0	-

$$2 - \frac{(-1) \cdot 2}{3} = 2 + \frac{2}{3} = \frac{8}{3} = 2\frac{2}{3}$$

$$-1 - \frac{-7 \cdot 2}{11} = -1 + \frac{14}{11} = \frac{3}{11}$$

$$5 - \frac{2 \cdot 2}{3} = \frac{15}{3} - \frac{4}{3} = \frac{11}{3}$$

$$4 - \frac{2 \cdot -1}{3} = \frac{12}{3} + \frac{2}{3} = \frac{14}{3}$$

$$-9 - \frac{2 \cdot -5}{3} = -\frac{21}{3} + \frac{10}{3} = -\frac{11}{3}$$

$$1 - \frac{-1 \cdot -1}{3} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$0 - \frac{-1 \cdot 2}{3} = 0 + \frac{2}{3} = \frac{2}{3}$$

$$3 - \frac{-1 \cdot -1}{3} = 3 + \frac{1}{3} = \frac{10}{3}$$

$$1 - \frac{-1 \cdot -5}{3} = 1 - \frac{5}{3} = \frac{3}{3} - \frac{5}{3} = -\frac{2}{3}$$

$$2 - \frac{-1 \cdot -7}{3} = 2 - \frac{7}{3} = \frac{6}{3} - \frac{7}{3} = -\frac{1}{3}$$

$$5 - \frac{-7 \cdot -9}{3} = 5 - \frac{63}{3} = \frac{15}{3} - \frac{63}{3} = -\frac{48}{3} = -16$$

Таблица 4:

И.Л.		x_1	x_2	x_3	x_4	x_5	x_6	x_7	СВ	
x_1	3	1	$\frac{1}{11}$	0	0				$\frac{33}{11}$	4,5
x_4	3	0	$\frac{1}{11}$	0	1				33	33
x_2	4	0	$\frac{2}{11}$	1	0				$\frac{11}{2}$	5,5
x_3	2	0	$\frac{4}{33}$	0	0					
x_5	0	0	0	0	0	1				

$$Z = 3x_1 + 0x_2 + 1x_3 + 3x_4$$

$$Z_{max} = Z(3, 0, 1, 3) =$$

$$= 3 \cdot 3 + 0 + 1 + 3 = 6 + 4 = 10$$

$$\text{Ответ: } Z_{max} = Z(5, 0, 1, 3) = 2$$

$\sqrt{4}$

$$Z = x_1 + x_2 - x_3 - 2x_4 \rightarrow \min$$

$$\begin{cases} x_1 - 2x_2 + x_4 = -3 \quad (*) \\ x_3 - 2x_4 = 2 \\ 3x_2 - x_4 \leq 5 \end{cases} \Rightarrow \begin{cases} -x_1 + 2x_2 - x_4 = 3 \\ x_3 - 2x_4 = 2 \\ 3x_2 - x_4 + x_5 = 5 \end{cases}$$

$$x_1, x_2, x_3, x_4 \geq 0$$

$$-Z = -x_1 - x_2 + x_3 + 2x_4 - M(x_6)$$

$$\begin{cases} -x_1 + 2x_2 - x_4 + x_6 = 3 \\ x_3 - 2x_4 = 2 \\ 3x_2 - x_4 + x_5 = 5 \end{cases}$$

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b \\ -1 & 2 & 0 & -1 & 0 & 1 & 3 \\ 0 & 0 & 1 & -2 & 0 & 0 & 2 \\ 0 & 3 & 0 & -1 & 1 & 0 & 5 \end{pmatrix}$$

$$\begin{cases} x_3 = 2 + 2x_4 \\ x_5 = 5 + x_4 - 3x_2 \\ x_6 = 3 + x_4 - 2x_2 + x_1 \end{cases}$$

$$-Z = -x_1 - x_2 + (2 + 2x_4) + 2x_4 - M(3 + x_4 - 2x_2 + x_1) \rightarrow \max$$

$$-Z = 2 - x_1 - x_2 + 4x_4 - M(3 + x_4 - 2x_2 + x_1) \rightarrow \max$$

Iteration 1

		x_1	x_2	x_3	x_4	x_5	x_6	CO
x_3	2	0	0	1	-2	0	0	-
x_5	5	0	3	0	-1	1	0	$\frac{5}{3}$
x_6	3	-1	2	0	-1	0	1	$\frac{3}{2}$
$-Z$	2	1	1	0	-4	0	0	Min
$-M$	-3	1	-2	0	1	0	0	Min

$\frac{10}{3}$
 $\frac{5}{2}$

$$-Z = -Z(0; 0; 2; 0; 5; 3) = -0 - 0 + 2 - M(3 + 0 - 2 \cdot 0 + 0) = 2 - 3M$$

Iteration 2:

		x_1	x_2	x_3	x_4	x_5	x_6	CO
C_j	1				-2	0		
x_2	2	0	0	1	$-\frac{1}{2}$	1		1
x_5	$\frac{1}{2}$	$\frac{3}{2}$	0	0	$-\frac{1}{2}$	0		
x_3	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	$-\frac{1}{2}$	0		
$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$	0	0	$-\frac{7}{2}$	0		
M	0	0	0	0	0	0		

$$2 - \frac{0 \cdot 3}{2} = 2; \quad 5 - \frac{3 \cdot 3}{2} = 5 - \frac{9}{2} = \frac{1}{2}$$

$$0 - \frac{(3 \cdot (-1))}{2} = \frac{3}{2}; \quad 0 - \frac{3 \cdot 1}{2} = -\frac{3}{2}$$

$$-1 - \frac{(3 \cdot (-1))}{2} = -1 + \frac{3}{2} = -\frac{2}{2} + \frac{3}{2} = \frac{1}{2}$$

$$2 - \frac{3 \cdot 1}{2} = \frac{4}{2} - \frac{3}{2} = \frac{1}{2}$$

$$1 - \frac{((-1) \cdot 1)}{2} = 1 + \frac{1}{2} = \frac{3}{2}$$

$$-4 - \frac{(-1 \cdot 1)}{2} = -4 + \frac{1}{2} = -\frac{8}{2} + \frac{1}{2} = -\frac{7}{2}$$

$$0 - \frac{1 \cdot 1}{2} = -\frac{1}{2}; \quad 3 - \frac{3 \cdot (-2)}{2} = -3 + 3 = 0$$

$$1 - \frac{(1 \cdot 1) \cdot (-2)}{2} = 1 - 1 = 0$$

$$1 - \frac{((-1) \cdot (-2))}{2} = 1 - 1 = 0$$

$$0 - \frac{(1 \cdot (-2))}{2} = 1$$

$$-2 = \frac{1}{2} (0; \frac{3}{2}; 2; 0; \frac{1}{2}; 0) =$$

$$= 2 - 0 - \frac{3}{2} + 4 \cdot 0 - M(3 + 0 - 2 \cdot \frac{3}{2} + 0) =$$

$$\frac{1}{2} - 0M = \frac{1}{2}$$

$$-M(3 + 0 - 2 \cdot \frac{3}{2} + 0) =$$

Maßstab 3:

$\downarrow$	1	x_1	x_2	x_3	x_4	x_5	0
x_3	4	6	0	1	0	4	
x_4	1	3	0	0	1	2	
x_2	2	1	1	0	0	1	
$-z$	4	10	0	0	0	7	

$$z - \left(\frac{(+2) \cdot \frac{1}{2}}{\frac{1}{2}} \right) = z + 2 = 4; \quad \frac{z}{2} - \left(\frac{\frac{1}{2} \cdot (-\frac{1}{2})}{\frac{1}{2}} \right) = \frac{z}{2} + \frac{1}{2} = 2$$

$$0 - \left(\frac{\frac{3}{2} \cdot (-2)}{\frac{1}{2}} \right) = 6; \quad 0 - \left(\frac{-2 \cdot 1}{\frac{1}{2}} \right) = 4$$

$$-\frac{1}{2} - \left(\frac{\frac{3}{2} \cdot (-\frac{1}{2})}{\frac{1}{2}} \right) = -\frac{1}{2} + \frac{3}{2} = 1$$

$$0 - \left(\frac{1 \cdot (-\frac{1}{2})}{\frac{1}{2}} \right) = 1; \quad \frac{z}{2} - \left(\frac{\frac{3}{2} \cdot (-\frac{7}{2})}{\frac{1}{2}} \right) = \frac{z}{2} + \frac{21}{2} = \frac{2z}{2} + \frac{21}{2} = \frac{2z+21}{2} = 12$$

$$0 - \left(\frac{1 \cdot (-\frac{7}{2})}{\frac{1}{2}} \right) = 7$$

$$-z_{\max} = -z(0; 2; 4; 1; 0) =$$

$$= -0 - 2 + 4 + 2 \cdot 1 = 4 =$$

$$\Rightarrow z_{\min} = z(0; 2; 4; 1; 0) = -4$$

Ordnung: $z_{\min} = z(0; 2; 4; 1; 0) = -4$

Линейное задание №4

$$Z = -x_1 - 4x_2 - x_3 \rightarrow \max$$

$$Z = x_1 + 4x_2 + x_3 \rightarrow \min$$

$$\begin{cases} 5x_1 + 12x_2 + 2x_3 + x_4 = 9 \\ 3x_1 + 4x_2 + 4x_3 + x_5 = 11 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases} \Rightarrow \begin{cases} 5x_1 + 12x_2 + 2x_3 + x_4 = 9 \\ 3x_1 + 4x_2 + 4x_3 + x_5 = 11 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$-Z = -x_1 - 4x_2 - x_3 - M(x_4 + x_5)$$

$$\begin{cases} x_4 = 9 - 5x_1 - 12x_2 - 2x_3 \\ x_5 = 11 - 3x_1 - 4x_2 - 4x_3 \end{cases}$$

$$-Z = -x_1 - 4x_2 - x_3 - M(20 - 8x_1 - 16x_2 - 6x_3)$$

Шаг 1:

б.л.		x_1	x_2	x_3	x_4	x_5	CO
x_4	9	5	12	2	1	0	$\frac{3}{4}$
x_5	11	3	4	4	0	1	$\frac{11}{4}$
$-Z$	0	1	4	1	0	0	20
$-M$	-20	-8	-16	-6	0	0	20

$$\begin{aligned} & 5 \cdot \frac{3}{4} = \frac{15}{4} \\ & 12 \cdot \frac{3}{4} = 9 \\ & 2 \cdot \frac{3}{4} = \frac{3}{2} \\ & 1 \cdot \frac{3}{4} = \frac{3}{4} \end{aligned}$$

Шаг 2:

б.л.		x_1	x_2	x_3	x_4	x_5	CO
x_2	$\frac{3}{4}$	$\frac{5}{12}$	1	$\frac{1}{6}$		0	4.5
x_5	8	$\frac{1}{3}$	0	$\frac{10}{3}$		1	2.4
$-Z$	-3	$-\frac{2}{3}$	0	$\frac{1}{3}$		0	
$-M$	-8	$-\frac{4}{3}$	0	$-\frac{10}{3}$		0	

$$\begin{aligned} & 3 \cdot \frac{3}{4} = \frac{9}{4} \\ & -8 \cdot \frac{3}{4} = -6 \\ & \frac{9}{4} - 6 = -\frac{15}{4} \end{aligned}$$

$$11 - \frac{9}{12} \cdot 4 = 11 - \frac{36}{12} = 11 - 3 = 8$$

$$0 - \frac{9}{12} \cdot 4 = -3 - 20 - \frac{3}{4} \cdot (-1) = -20 + \frac{3}{4} = -19.75$$

$$3 - \frac{5}{12} \cdot 4 = \frac{36}{12} - \frac{20}{12} = \frac{16}{12} = \frac{4}{3}$$

$$1 - \frac{5 \cdot 4}{12} = \frac{12}{12} - \frac{20}{12} = \frac{-8}{12} = -\frac{2}{3}$$

$$-8 - (5 \cdot (-\frac{2}{3})) = -8 + \frac{20}{3} = \frac{-24 + 20}{3} = -\frac{4}{3}$$

X_3

$$4 - \frac{2 \cdot 4}{12} = 4 - \frac{8}{12} = 4 - \frac{2}{3} = \frac{12 - 2}{3} = \frac{10}{3}$$

$$1 - \frac{2 \cdot 4}{12} = 1 - \frac{2}{3} = \frac{1}{3}$$

$$-6 - (2 \cdot (-\frac{2}{3})) = -6 + \frac{8}{3} = \frac{-18 + 8}{3} = -\frac{10}{3}$$

X_2

$$\frac{8}{4} : \frac{1}{8} = \frac{3}{4} \cdot 8 = \frac{18}{4} = 4.5$$

$$8 : \frac{10}{3} = 8 \cdot \frac{3}{10} = \frac{24}{10} = 2.4$$

Максимум 3:

б.н.	1	X_1	X_2	X_3	X_4	X_5	б.
X_2	$\frac{2 \cdot 12}{20 \cdot 80}$	$\frac{2}{20}$	1	0			12
X_3	$\frac{12}{5}$	$\frac{2}{5}$	0	1			6
$-\frac{2}{3}$	$-\frac{19}{5}$	$-\frac{12}{15}$	0	0			12
-11	0	0	0	0			12

$$\frac{4}{3} \cdot \frac{8}{10} = \frac{4}{10} = \frac{2}{5}$$

$$\frac{3}{4} - \frac{1}{6} \cdot 8 = \frac{10}{4}$$

$$\frac{5}{12} - \frac{1}{6} \cdot \frac{4}{3} = \frac{5}{12} - \frac{4}{18} = \frac{5}{12} - \frac{2}{9} = \frac{15 - 8}{36} = \frac{7}{36}$$

$$\frac{3}{4} - \frac{8}{3} \cdot \frac{5}{5} = \frac{3}{4} - \frac{4}{15} = \frac{45 - 16}{60} = \frac{29}{60}$$

$$-3 - 8 \cdot \frac{1}{3} = -3 - \frac{8}{3} = -\frac{38}{3} = -\frac{19}{1.5}$$

$$-8 - 8 \cdot \frac{10}{3} = -8 - \frac{80}{3} = -\frac{96 + 80}{3} = -\frac{176}{3}$$

$$-\frac{2}{3} - \frac{4}{3} \cdot \frac{1}{3} = -\frac{2}{3} - \frac{4}{9} = -\frac{6 + 4}{9} = -\frac{10}{9}$$

Task 4:

b_i	1	x_1	x_2	x_3	10
x_1	1	1	$\frac{20}{7}$	0	
x_3	2	0	$-\frac{8}{7}$	1	
$-z$	-3	0	16	0	

$$\frac{12}{5} - \frac{\frac{8}{20} \cdot \frac{2}{5}}{\frac{2}{20}} = \frac{12}{5} = 2$$

$$-\frac{13}{5} - \left(\frac{\frac{8}{20} \cdot -\frac{12}{15}}{\frac{2}{20}} \right) = -\frac{13}{5} + \frac{12}{15} = -\frac{27}{15} = -\frac{9}{5}$$

$$-3 - 8 = \frac{1}{3} = -3 - \frac{8}{10} = -\frac{30}{10} - \frac{8}{10} = -\frac{38}{10}$$

$$0 - 1 \cdot \frac{2}{5} = -\frac{2}{5} - \frac{80}{7} = -\frac{8}{7}$$

$$0 - 1 \cdot \frac{-12}{7} = \frac{12}{7} \cdot \frac{4}{3} = \frac{48}{7} = 16$$

$$-Z_{\max} = -1 - 4 \cdot 0 - 2 = -3 =$$

$$\Rightarrow Z_{\min} = Z(1; 0; 2) = 3$$

$$\text{Answer: } Z_{\min} = Z(1; 0; 2) = 3$$

$\sqrt{5}$

$$Z = -x_1 + 2x_2 - 3x_3 \rightarrow \max$$

$$\begin{cases} -2x_1 + x_2 + 3x_3 = 2 \\ 2x_1 + 3x_2 + 4x_3 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{cases} -2x_1 + x_2 + 3x_3 = 2 \\ 2x_1 + 3x_2 + 4x_3 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{pmatrix} -2 & 1 & 3 & 0 & 0 \\ 2 & 3 & 4 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{cases} -2x_1 + x_2 + 3x_3 + x_4 = 2 \\ 2x_1 + 3x_2 + 4x_3 + x_5 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{cases} -2x_1 + x_2 + 3x_3 + x_4 = 2 \\ 2x_1 + 3x_2 + 4x_3 + x_5 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{cases} -2x_1 + x_2 + 3x_3 + x_4 = 2 \\ 2x_1 + 3x_2 + 4x_3 + x_5 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$Z = -x_1 + 2x_2 - 3x_3 - M(x_4 + x_5)$$

$$\begin{cases} x_4 = 2 + 2x_1 - x_2 - 3x_3 \\ x_5 = 1 - 2x_1 - 3x_2 - 4x_3 \end{cases}$$

$$\begin{cases} x_4 = 2 + 2x_1 - x_2 - 3x_3 \\ x_5 = 1 - 2x_1 - 3x_2 - 4x_3 \end{cases}$$

$$Z = -x_1 + 2x_2 - 3x_3 - M(3 - 4x_2 - 7x_3)$$

Iteration 1:

	B.R.	1	x_1	x_2	x_3	x_4	x_5	(0)
x_4	2	-2	1	3	1	0	0	$\frac{2}{3}$
x_5	1	2	3	4	0	1	0	$\frac{1}{4}$
Z	0	1	-2	3	0	0	0	
$-M$	-3	0	-4	-7	0	0	0	

$$Z = Z(0; 0; 0; 2; 1) =$$

$$= -0 + 2 \cdot 0 - 3 \cdot 0 - M(3 - 4 \cdot 0 - 7 \cdot 0) = 0 - 3M$$

Шаг 2:

		x_1	x_2	x_3	x_4	x_5	CO
f.n	1	$-\frac{11}{4}$	$-\frac{5}{4}$	0	1		
x_1	$\frac{5}{4}$	1	$\frac{3}{4}$	1	0		
x_3	2	$\frac{1}{2}$	$\frac{1}{4}$	0	0		
2	$-\frac{3}{4}$	$-\frac{1}{2}$	$-\frac{17}{4}$	0	0		
-11	$-\frac{5}{4}$	$\frac{11}{4}$	$\frac{17}{4}$	0	0		

$$0 - \frac{8}{4} \cdot 1 = -\frac{8}{4} = -2$$

$$-2 - \frac{3}{4} \cdot 2 = -2 - \frac{6}{4} = -\frac{8}{4} - \frac{6}{4} = -\frac{14}{4}$$

$$1 - \frac{3}{4} \cdot 3 = 1 - \frac{9}{4} = \frac{4}{4} - \frac{9}{4} = -\frac{5}{4}$$

$$0 - \frac{1}{4} \cdot 3 = -\frac{3}{4}$$

$$-3 - \frac{(1 \cdot -7)}{4} = -3 + \frac{7}{4} = -\frac{12}{4} + \frac{7}{4} = -\frac{5}{4}$$

$$1 - \frac{2 \cdot 3}{2} = \frac{2}{2} - \frac{6}{2} = -\frac{4}{2} = -2$$

$$-2 - \frac{3}{4} \cdot 5 = -2 - \frac{15}{4} = -\frac{8}{4} - \frac{15}{4} = -\frac{23}{4}$$

$$0 - 2 \cdot -7 = \frac{14}{4}$$

$$-4 - \frac{3}{4} \cdot -7 = -4 + \frac{21}{4} = \frac{-16}{4} + \frac{21}{4} = \frac{5}{4}$$

Остатки не вычисляются